Oct 21 — What we did today

IAA AI-Assisted Hackathon 2026 · Woojin Ji & Krista Stanke · Part 1 closes Friday, Oct 23

> **In one line.** We moved the model from notebook cells into a **pipeline that rebuilds the submission with one command**, and added **flood-geography interactions** to the buildings GLM for our best score (**v11, 0.12307**). Blending in a GBM scores higher but over-prices the safest customers, so we did not use it; that trade-off goes to Thursday's team decision.

1. Score journey

Version	Change	Public score	In plain words
v9 (Oct 20)	Motor frequency shifts removed	0.12274	Yesterday's adopted model
v10	Lynden-Bell completeness + relative premium by start month	0.12265	Corrections adopted by rule; within the expected range (−0.00009)
v11 (adopted)	**Five flood-geography interactions in the buildings GLM**	**0.12307**	**Best score so far**

> **Rank.** 23rd of 39 on the public board. Most of the gap to the top (0.125–0.130) is likely whether a GBM is blended into buildings (section 5). Prizes depend on the private scores of Parts 1 and 2 plus the panel's assessment, not on the Part 1 public rank (Rules 2.4).

2. What we did

Each item: Why · Found · Decided

① A reproducible pipeline

Why A notebook remembers the state of earlier cells, so the same code can give different results depending on the order run. Winners must also hand over code that rebuilds the submission (Rules 2.9).

Found `python -m src.pipeline` rebuilt v9 **exactly** (0.000% difference, correlation 1.000000); the Pareto loading of 2.83 came out of the code again.

Decided Every number in one configuration file (`src/config.py`). Tests 32 → 46, tests run on every push (CI), package versions recorded.

② A Day 1 checks tool, and a 2021 rate increase

Why To see what is different about the Part 2 data with one command as soon as it arrives.

Found All known steps (own damage, BI) and the leaking field were found (positive controls passed). New: a **+50–60% step in premium in May–June 2021**, followed by more lapses and, from 2022, more new business.

Decided Lower-risk customers leaving after the increase (adverse selection) is recorded as a **hypothesis** only.

③ Fixing the relative premium (v10)

Why Because of the 2021 rate step, dividing premium by the year's average made early-2021 policies look cheap and later ones dear: a price change read as risk.

Found Dividing by the start month's average: contents D^2 +0.0003, tails within limits.

Decided Adopted, together with Lynden-Bell completeness, as v10. **Our own checks tool caught a flaw the AI had built the day before.**

④ Flood interactions in buildings (v11, the main result)
Why The GBM gains +0.009 D^2 in buildings but over-prices the safest customers. Could part of that gain be moved into a GLM that still reads as rating tables?
Found Pairs the GBM uses together, added as one relativity per combination: **D^2 +0.0025** (28% of the GBM's gain), tails slightly better. All five pairs are **flood geography**: within a flood zone, height and distance from the river change the risk a lot, and larger houses lose more.
Decided Adopted (v11). The validation years were never used to choose the pairs.

3. Tried, not adopted

Experiment	Result	Why not
Motor interactions	D^2 +0.0001	Driver-experience pairs unstable between years; main effects are enough for motor
Buildings GBM blend (75% GBM)	D^2 +0.0072	Safest 10% **over-priced by about 27%** → fails the tail rule. Team decision on Thursday

4. Mistakes and lessons

- **Notebook kernel:** the AI suggested running the blend in a notebook; the kernel still held old `src` code, so the first run compared against the wrong GLM. Caught because the baseline matched E7's main-effects row to four decimals; re-run in the terminal. → experiments that import `src` run from the terminal.
- **Relative premium:** the AI's design flaw from Oct 20, found by our checks tool.
- **v11 prediction:** expected +0.0005 to +0.0012, observed +0.00042 (right direction, just below). Leaderboard predictions: 9 of 12 within range.

5. Thursday's team decision: score vs pricing quality

Buildings	v11 GLM + interactions	Blend (75% GBM)
Validation D^2 (average)	0.0677	0.0749 (+0.0072)
Safest 10% A/E (2023 / 2024)	1.147 / 0.901	0.735 / 0.720
Riskiest 10% A/E	1.017 / 1.083	1.118 / 1.145
Expected leaderboard effect	—	+0.002 to +0.003

> D^2 barely penalises over-pricing the safest customers, because their expected cost is small. The rule and our recommendation is v11, consistent with the report's thesis (transparent models, trade-offs measured). If the team changes the rule, the change is recorded openly with its reasons.

6. What is left

- **Thursday:** figure script (before the call) · two team decisions (blend, motor p = 1.4) · fix the final model (currently v11) · 7 pm Zoom · (optional) customers with 2+ past claims
- **Friday:** select the **one final submission by hand** on Kaggle (otherwise the best public score is picked automatically)